

Project Design Phase

Proposed Solution Template

Date	28 June 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

S.No.	Parameter	Description
1.	Problem Statement (Problem to be solved)	Farmers lack a simple, data-driven tool to choose the most suitable crop based on their soil nutrients and climate conditions, leading to poor yield and financial loss.
2.	Idea / Solution Description	A web-based ML application that accepts 7 soil and climate inputs (N, P, K, temperature, humidity, pH, rainfall) and recommends the best-suited crop with a confidence score and top-3 alternatives.
3.	Novelty / Uniqueness	Combines a trained Random Forest classification model achieving 99%+ accuracy with a zero-installation Streamlit web interface accessible to non-technical users including rural farmers.
4.	Social Impact / Customer Satisfaction	Reduces financial risk for farmers, improves crop yield, supports better utilization of water and fertilizer resources, and empowers farmers to make scientific decisions independently.

5.	Business Model (Revenue Model)	Can be offered free as a public service through agricultural extension programs, or as a freemium product with premium features such as SMS crop alerts and fertilizer recommendations.
6.	Scalability of the Solution	Can be extended to include fertilizer recommendations, real-time weather API integration, regional language support (Telugu, Hindi), and a mobile app for wider rural accessibility.